TRACE-ESUS in Plain English

A clinician-friendly guide to the preliminary results and the three R01 aims

For clinical collaborators and trainees who do not work in causal AI or latent-variable modeling.

The 30-second explanation

Patients with ESUS can have the same abnormal biomarker for very different biological reasons. TRACE-ESUS is designed to separate a likely stroke-related atrial mechanism from look-alike explanations such as kidney disease, heart failure, inflammation, or competing vascular disease. Aim 1 tests when the method works. Aim 2 asks whether the same biological pattern appears in real patients across two cohorts. Aim 3 tests whether the locked score remains reliable in later patients and can be generated safely in a live clinical workflow.

What this guide explains

1. Why the clinical problem is difficult.
2. What the current experiments actually show - and what they do not show.
3. What each aim will do, how it will be tested, and what it will produce.
4. Plain-English meanings of counterfactual, fingerprint, recalibration, freezing, and abstention.

Important boundary

All currently verified numerical performance results in this guide come from synthetic experiments, where the true mechanism is known by design. Real ARCADIA analyses are ongoing. In real patients, we can test stability and biological consistency, but we cannot honestly claim to know the hidden true mechanism for every individual.

Source basis

Current TRACE-ESUS Specific Aims page and the 3 August 2026 technical update.

1. The clinical problem: the same clue can come from different causes

A biomarker is a clue. It is not automatically the cause of the stroke.

Patient A: true atrial disease	Patient B: kidney-related look-alike
NT-proBNP is high because atrial or cardiac disease is producing the signal. An atrial-thromboembolic mechanism may be biologically plausible.	NT-proBNP is also high, but kidney dysfunction is contributing strongly. The patient can look "atrial" even when the stroke-relevant mechanism is different.

Why a conventional pattern model can be misled

It may ask, "Which known patient profile does this person resemble?" If high NT-proBNP is common in the atrial-looking group, both patients may be assigned there.

What TRACE-ESUS asks instead

- Which biological mechanism could have generated the evidence?
- Could kidney disease, heart failure, inflammation, or vascular disease explain the same evidence?
- If the proposed atrial mechanism were switched off, would the atrial evidence disappear?
- Is there enough information, or should the model abstain?

Simple counterfactual example

A patient has high NT-proBNP, an abnormal atrial electrical marker, and kidney disease. A resemblance question asks, "Which group does this look like?" A counterfactual question asks, "If the atrial mechanism were absent, could kidney disease alone still explain the pattern?" If yes, confidence in an atrial mechanism should fall.

The model is not selecting a drug

The near-term goal is mechanism characterization and diagnostic prioritization. The system should communicate the likely mechanism, uncertainty, missing information, and when more testing is needed - not recommend anticoagulation.

2. Preliminary studies: what the current results mean

First, remember the setting

The computer generated each simulated patient and secretly recorded the true mechanism. The models did not see the answer. This is why true mechanism-recovery accuracy can be measured here.

Result 1: kidney-related biomarker distortion

Question: When kidney disease raises an NT-proBNP-like marker, does the model incorrectly call the patient atrial?

Model	Accuracy	False atrial
Pooled association	57.9%	76.0%
Renal-adjusted association	81.6%	19.7%
Biology-constrained causal model	81.9%	18.5%
Oracle ceiling	82.7%	17.3%

Plain-English interpretation

Among 100 kidney-impaired patients whose true simulated mechanism is competing rather than atrial, the pooled model would wrongly call about 76 atrial. Once kidney effects are modeled explicitly, the error falls to about 19. The causal and correctly renal-adjusted models perform almost the same.

What this supports

- Kidney effects must be represented explicitly.
- Ignoring a nuisance cause can create many false atrial assignments.

What it does not yet prove

- It does not prove counterfactual superiority over every strong adjusted model.
- The causal model was only 0.30 percentage points better than the renal-adjusted comparator here.

Result 2: the K=1 null test

Question: If there is only one true group, will the method invent two endotypes?

Model	False selection of two groups
Pooled association	500 of 500 runs
Renal-adjusted association	0 of 500
Biology-constrained causal model	0 of 500

Meaning

The pooled model mistook kidney-related variation for a biological subtype. The careful conclusion is: unmodeled nuisance structure can create false endotypes.

3. Preliminary studies continued: what happens in a new hospital

A model may fail elsewhere because patient mix, kidney disease prevalence, laboratory practice, or missing-data patterns change.

Result 3A: isolated kidney shift

Model	Accuracy lost
Pooled association	3.90 percentage points
Modular causal model	0.03 percentage points

Meaning

The modular model kept the stable biological part and adjusted the kidney-related part. It was almost unaffected by the kidney-only change.

Result 3B: several hospital differences occur together

Model under strong shift	Accuracy	False atrial
Pooled association	73.4%	44.7%
Frozen causal model	76.2%	36.6%
Target-calibrated association	77.1%	23.4%
Modular causal model	77.6%	22.5%
Oracle ceiling	78.1%	22.2%

Meaning

A causal graph is not automatically transportable. The frozen causal model still made many false atrial calls. Performance improved only after limited recalibration of the changeable nuisance and measurement parts. The modular model came close to the oracle, but only slightly exceeded target-calibrated association.

The honest overall conclusion

- Explicit nuisance modeling is essential.
- Separating stable biology from site-specific nuisance can help transport.
- A frozen causal graph alone is insufficient.
- Missing information may require abstention.
- Counterfactual value still needs a decisive same-model test.

The decisive next experiment

Fit one causal model once. Ask two questions of that exact model: posterior resemblance and counterfactual sufficiency/disablement. Any difference can then be attributed to the query, not to a different model.

4. Why the project uses several kinds of data

No single dataset can answer every question. TRACE-ESUS uses an evidence ladder.

Data level	What it provides	What can be measured honestly
Fully synthetic	Complete control of hidden mechanisms and nuisance effects.	True mechanism accuracy and exact error sources.
ARCADIA-calibrated semi-synthetic	Realistic distributions and missingness, but a known hidden answer.	Realistic stress tests with known truth.
Real ARCADIA	A trial cohort enriched for atrial-cardiopathy markers.	Coverage, stability, biological coherence, sensitivity - not true-mechanism accuracy.
Brown/Rhode Island	An independent real-world cohort.	Independent replication and held-out biological validation.
Later Brown/Rhode Island patients	Patients after a fixed cutoff, never used for development.	Temporal robustness, drift, calibration, and silent workflow feasibility.

What "ARCADIA-calibrated semi-synthetic" means

Blueprint analogy

ARCADIA provides realistic ranges, relationships, prevalence, and missingness. The simulator builds new patients that resemble that blueprint but also records each hidden mechanism. This combines realism with an answer key.

Why real ARCADIA cannot provide true-mechanism accuracy

No definitive test reveals the complete hidden ESUS mechanism in every real patient. Expert judgment is valuable but is not ground truth. Real-data validation must therefore use reproducibility, held-out biology, negative controls, and sensitivity analyses.

What "outcome-blind" means

Treatment assignment and recurrent-stroke outcomes are not used while constructing the score. This prevents the team from designing a score that appears favorable only because it was shaped around the outcome.

1 Determine when the method works - and when it should not be trusted

Scientific question

When does modular causal endotyping add value beyond simpler adjusted methods, and when is the evidence too ambiguous?

What the team will do

- Create synthetic and ARCADIA-like semi-synthetic patients with known mechanisms.
- Compare clinical rules, adjusted association, target calibration, modular causal modeling, counterfactual querying, uncertainty, and abstention.
- Stress-test overlap, kidney/heart-failure/inflammatory distortion, missingness, graph error, and distribution shift.
- Compare posterior and counterfactual questions using the same fitted model.

Three acceptable answers

A simple adjusted method is sufficient; the modular causal method adds useful information; or the evidence is non-identifying and the model should abstain. Aim 1 does not force the causal model to win everywhere.

How it will be evaluated

Evaluation	Plain-English meaning
Mechanism recovery	Does it identify the hidden mechanism?
False atrial attribution	How often does it wrongly call a case atrial?
K=1 false discovery	Does it invent groups when only one exists?
Calibration	Does stated confidence match actual correctness?
Uncertainty/coverage	Are uncertainty ranges reliable?
Selective risk	How accurate is it after uncertain cases are withheld?
Transport degradation	How much performance is lost after a shift?

What success looks like

- Prespecified settings show where each component adds value.
- Simple settings do not suffer unnecessary loss.
- Null or non-identifiable settings trigger appropriate abstention.

Aim 1 product

A frozen method, open software, and an operating atlas explaining when to use a simple adjusted model, a modular causal model, or no mechanism claim.

2 Find and independently replicate biological mechanism patterns in real patients

Scientific question

Does an outcome-blind atrial-thromboembolic mechanism pattern appear reproducibly in ARCADIA and Brown/Rhode Island?

ARCADIA development

- Audit marker overlap and missingness.
- Fit staged models without treatment or recurrent-stroke outcomes.
- Test bootstrap/split-sample stability, biological controls, graph sensitivity, and missing-not-at-random assumptions.
- Create an ARCADIA development-locked continuous score.

Independent Brown/Rhode Island replication

- Fit independently rather than copy ARCADIA labels.
- Align results using electrical, structural, biochemical, renal/heart-failure, and vascular anchors.
- Test held-out AF, ECG, echocardiography, rhythm monitoring, imaging, or recurrent-stroke evidence.

What is a mechanism fingerprint?

Fingerprint analogy

It is the full biological pattern, not one biomarker: which markers rise, how strongly, what happens when a mechanism is hypothetically disabled, and whether held-out clinical evidence agrees. Cohorts need not have identical numbers, but the meaningful pattern should be equivalent within prespecified margins.

Fallbacks that avoid forcing a result

- If discrete classes are unstable, use continuous mechanism axes.
- If marker overlap is incomplete, test a shared invariant core with site-specific measurement modules.

Aim 2 product

A cross-cohort-supported mechanism atlas and a final frozen, outcome-independent score with calibrated uncertainty.

3 Test later patients, silent clinical operation, and readiness for a future trial

Scientific question

After the score is frozen, does it remain reliable in later patients, identify unsafe use, and run in a real workflow?

Temporal holdout validation

- Choose the cutoff date before analysis.
- Use later Brown/Rhode Island patients only for Aim 3.
- Compare zero-shot use, constrained nuisance/measurement recalibration, and the strongest associative benchmark while locking the biological module.

What recalibration means

Thermometer analogy

The biology is body temperature; the hospital measurement system is the thermometer. Recalibration corrects the measurement scale without redefining fever. TRACE-ESUS can adjust nuisance and measurement modules, but not redesign the biological core after seeing later patients.

Prospective silent pilot

The score runs in the background without being shown to clinicians or changing care. It tests data availability, time to score, missingness, uncertainty, abstention, and pipeline reliability.

What abstention means

Clinical example

A patient has elevated NT-proBNP, severe renal dysfunction, no usable electrical marker, and missing echocardiography. The safe response may be: "Evidence is insufficient; obtain more information or do not use this score for mechanism classification."

Future trial design

Score distributions, uncertainty, censoring, and recurrent-event rates will estimate the sample size and event requirements for a future mechanism-stratified trial. The R01 designs that trial; it does not prove treatment benefit.

Aim 3 product

Temporal validation, a calibrated-abstention framework, silent-feasibility evidence, and a simulation-justified enrichment design for a future trial.

5. How the three aims fit together

Aim	Main question	Why it comes first	Product
1	Can the method recover mechanisms when truth is known?	We must know its limits before real-patient claims.	Frozen method and operating atlas.
2	Does reproducible biological structure exist in real patients?	A stable signal is required before prospective use.	Replicated atlas and final score.
3	Does the locked score remain reliable later and work silently?	Prospective use must test drift, uncertainty, and workflow.	Validation, abstention, and trial design.

The central logic

Aim 1 determines when the method deserves trust; Aim 2 determines whether the biological signal exists and replicates; Aim 3 determines whether the locked score remains safe and usable in future patients.

What the proposal can responsibly claim

- Current synthetic results support explicit nuisance modeling and modular recalibration.
- The project will determine whether counterfactual querying adds incremental value.
- Real data will establish stability, biological coherence, and replication - not perfect knowledge of each hidden mechanism.
- The R01 prepares a future mechanism-guided trial; it does not prescribe anticoagulation.

What still must be confirmed before submission

- Verified ARCADIA coverage, pilot results, and a defensible real-data comparator endpoint.
- Confirmed Brown/Rhode Island sample size, marker overlap, follow-up, and access timeline.
- Prespecified temporal cutoff and silent-pilot sample size.
- Formal definitions of fingerprint equivalence, recalibration limits, and abstention.

Why clinical input is essential

Neurologists and cardiology collaborators must define plausible pathways, dangerous errors, held-out validation evidence, and the combinations of missing or conflicting information that should make the model say, "I cannot call this mechanism."

6. Plain-English glossary

Term	Meaning
Endotype	A subgroup or continuous pattern defined by an underlying biological mechanism, not only outward appearance.
Nuisance pathway	A process such as kidney disease that changes a biomarker but may not be the stroke-causing mechanism of interest.
Posterior inference	Which mechanism or group does this patient most resemble, given the observed data?
Counterfactual inference	Would the evidence remain if a proposed mechanism were hypothetically removed or activated?
Semi-synthetic data	Computer-generated patients calibrated to resemble a real cohort while retaining a known hidden answer.
Outcome-blind	Treatment and recurrent-stroke outcomes are not used while constructing or freezing the score.
Freeze/lock	The rules are finalized before later data or outcomes are examined.
Mechanism fingerprint	The multivariable biological pattern expected from a mechanism across markers and held-out evidence.
Transportability	The score remains meaningful in another cohort, site, or time period.
Recalibration	Limited adjustment of prevalence, measurement, or nuisance processes without redesigning the biological core.
Abstention	The model declines to classify when evidence is insufficient or outside its safe-use range.

Final message for a medical collaborator

TRACE-ESUS is not simply a more complicated prediction model. It is a disciplined process for separating plausible biology from look-alike biomarker patterns, testing whether the inferred biology replicates, and refusing to make a claim when the available evidence is inadequate.